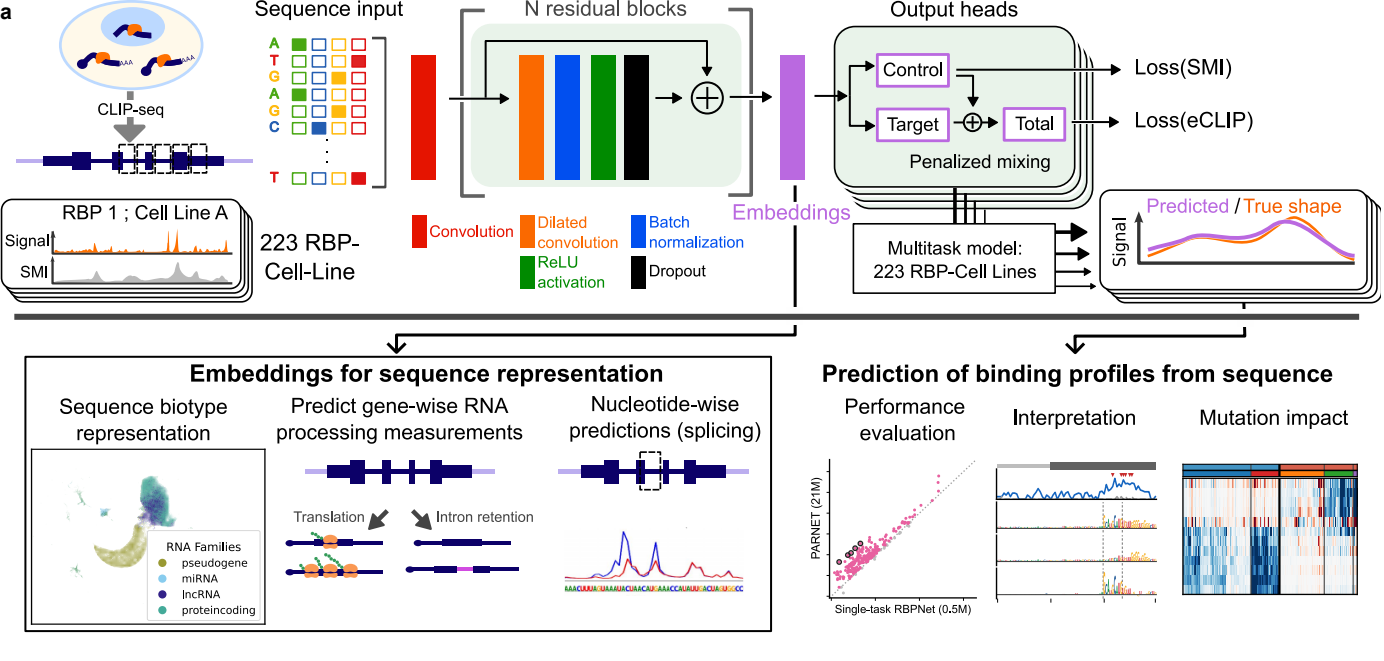
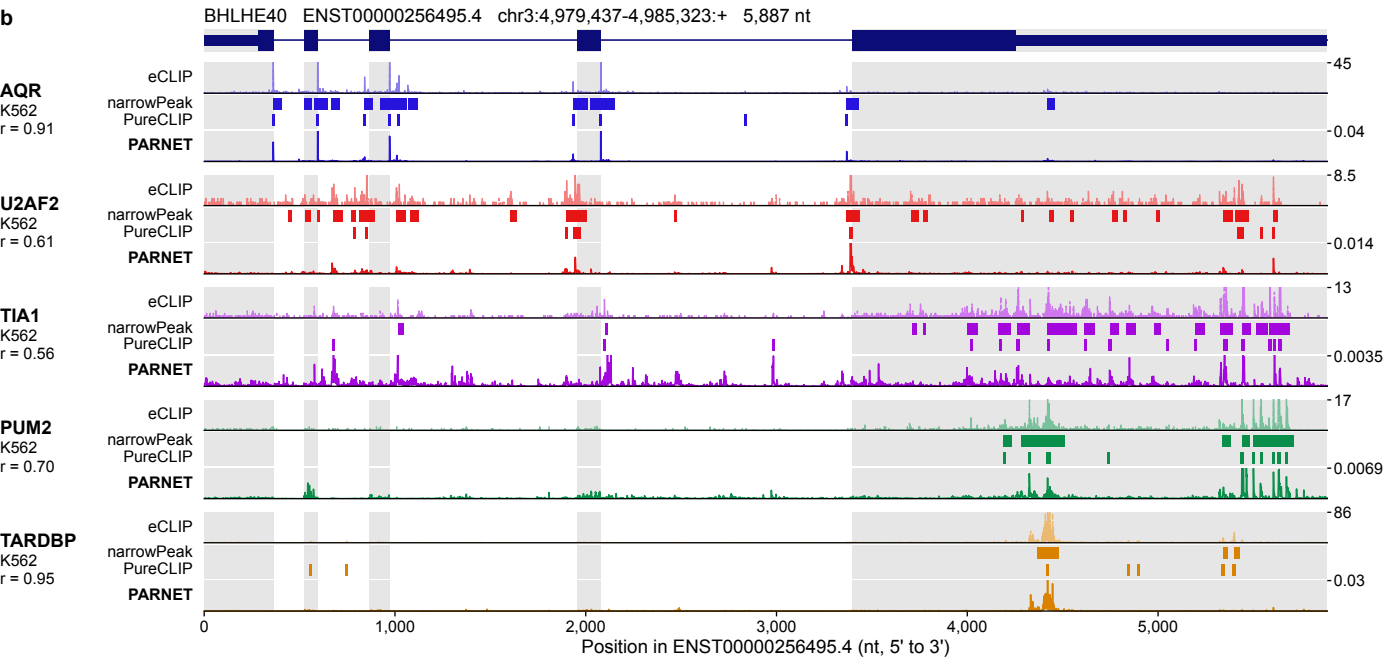

a
CLIP-seq
Sequence input
N residual blocks
Output heads
Control
Target
Total
Loss(SMI)
Loss(eCLIP)
Penalized mixing
Embeddings
RBP 1 ; Cell Line A
Signal
SMI
223 RBP-Cell-Line
Convolution
Dilated convolution
Batch normalization
ReLU activation
Dropout
Multitask model: 223 RBP-Cell Lines
Predicted / True shape
Signal
Embeddings for sequence representation
Sequence biotype representation
Predict gene-wise RNA processing measurements
Nucleotide-wise predictions (splicing)
RNA Families
pseudogene
miRNA
lncRNA
proteincoding
Translation
Intron retention
Prediction of binding profiles from sequence
Performance evaluation
Interpretation
Mutation impact
PARNET (21M)
Single-task RBPNet (0.5M)


b
BHLHE40 ENST00000256495.4 chr3:4,979,437-4,985,323:+ 5,887 nt
AQR
K562
r = 0.91
U2AF2
K562
r = 0.61
TIA1
K562
r = 0.56
PUM2
K562
r = 0.70
TARDBP
K562
r = 0.95
eCLIP
narrowPeak
PureCLIP
PARNET
45
0.04
8.5
0.014
13
0.0035
17
0.0069
86
0.03
0
1,000
2,000
3,000
4,000
5,000
Position in ENST00000256495.4 (nt, 5' to 3')